# Topology Optimization of Planar Elastic Energy-Storing Structures in a Fixed Area

Lab III – Project Proposal

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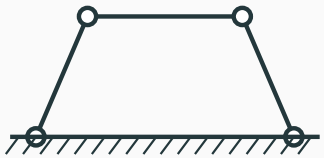
# Compliant vs. Traditional Mechanisms

**Traditional (rigid-body)**

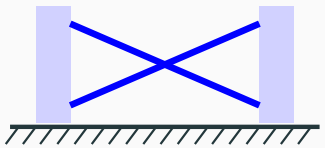
- Rigid links + discrete joints (pin, slider, bearing)
- Motion from assembled kinematic pairs
- Friction, backlash, wear; needs lubrication
- Multi-part assembly; energy stored externally (coil spring, torsion bar)

**Compliant (flexure-based)**

- Single monolithic part; motion = elastic deformation [1]
- No sliding contacts ⇒ no friction or wear
- Backlash-free, miniaturisable to MEMS [2]
- Energy stored *in* the structure itself



Pin-jointed linkage

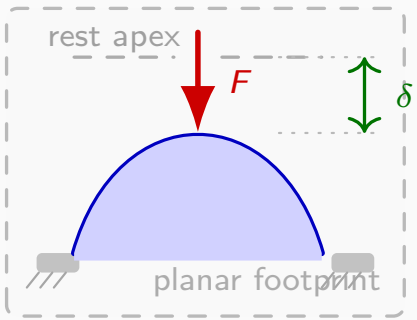


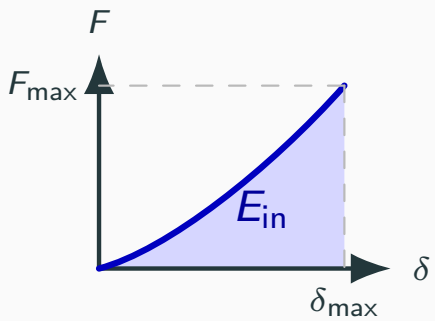
Compliant flexure pivot

# Background: Planar Elastic Energy Storage

- Compliant flexures store and release energy via reversible elastic deformation [1].
- Applications: MEMS actuators, soft grippers, watch springs, miniature energy harvesters [2].
- Often confined to thin planar footprints (PCB-mounted, wearable, additively manufactured) [3].
- Key metric — **energy storage efficiency**:

$$\eta = \frac{E_{\text{out}}}{E_{\text{in}}} = \frac{\int F_{\text{out}}\ \mathrm{d}x}{\int F_{\text{in}}\ \mathrm{d}x}$$



# Motivation & Research Gap

- Within a fixed planar footprint, geometry dictates the load path and strain distribution, hence the stored elastic energy at a given material volume [1].
- Folded / serpentine layouts lengthen the effective beam inside the same area, distributing strain more uniformly and delaying yield at the maximum-stress fibre.
- Topology optimisation (SIMP) generates non-intuitive material layouts that minimise compliance under a volume constraint [4], [5], [6].
- **Gap:** experimental comparisons of fold count and pattern family at *equal volume* and *equal footprint* are scarce; most topology-optimised planar springs are reported only numerically [2], [3].

# Project Goals

1. **Define objective function** for elastic energy storage performance within a fixed planar footprint ($100 \times 100 \times 3\,\mathrm{mm}$).
2. **Generate candidate structures** via:
   - Manual baseline designs (straight, folded, serpentine, curved beams)
   - Computational topology optimisation (SIMP method)
3. **Compare** force–displacement behaviour, energy storage $E_{\mathrm{in}}$, energy release $E_{\mathrm{out}}$, and efficiency $\eta$ across topologies.
4. **Analyse** how topology affects load paths, deformation modes, and stiffness distribution.

# Hypotheses

- **H1 — Fold-count effect within a pattern family:** For a fixed planar footprint ($100 \times 100 \times 3\,\mathrm{mm}$) and constant material volume, the elastic energy storage efficiency $\eta$ of a folded beam varies systematically with the *number of folds* within a given pattern (e.g. a triangular wave with 1, 2, 3, ... folds; a square wave across multiple folds).
- **H2 — Topology optimisation:** Topology-optimised structures will outperform manually designed baselines in terms of energy efficiency and performance stability.

# Experimental Variables

| | **H1: Fold count / pattern** | **H2: Topology optimisation** |
|---|---|---|
| **Independent** | Number of folds $n$<br>Pattern family (triangular wave, square wave, ...) | Design method (manual baseline vs. SIMP-optimised) |
| **Dependent** | $\eta$, $E_{\text{in}}$, $E_{\text{out}}$, stiffness $k$, peak force $F_{\text{max}}$ | $\eta$, $E_{\text{in}}$, $E_{\text{out}}$, stiffness $k$, trial-to-trial variance |
| **Controlled** | Material volume<br>Footprint $100\times100\times3$ mm<br>PLA + print parameters<br>Clamping BCs<br>Quasi-static loading rate | Volume fraction $f$<br>Footprint $100\times100\times3$ mm<br>PLA + print parameters<br>Clamping BCs<br>Quasi-static loading rate |

# Theory Framework

The theoretical basis is divided into three parts:

1. **Topology optimization model**: convert structure design into a material distribution problem.
2. **Mechanical constraints**: ensure the design is safe under static and cyclic loading.
3. **Energy-based objective**: optimise stored elastic energy and energy release efficiency.

# Topology Optimization Model

- The design domain is divided into finite elements.
- Each element has a density variable:

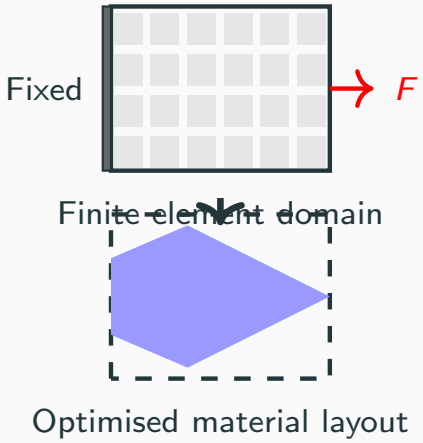
$$\rho_e \in [0, 1]$$

- $\rho_e = 1$ represents solid material, while $\rho_e = 0$ represents void.
- The SIMP method connects density and material stiffness [4], [5]:

$$E_e = E_{\min} + \rho_e^p (E_0 - E_{\min})$$

- The structural response is obtained from:

$$\mathbf{K}(\boldsymbol{\rho})\mathbf{U} = \mathbf{F}$$



# Mechanical and Physical Constraints

- The optimised structure must be mechanically feasible, not only numerically optimal.
- **Static strength**: avoid yielding or fracture:

$$\sigma_{\mathrm{vm}} \leq \sigma_y$$

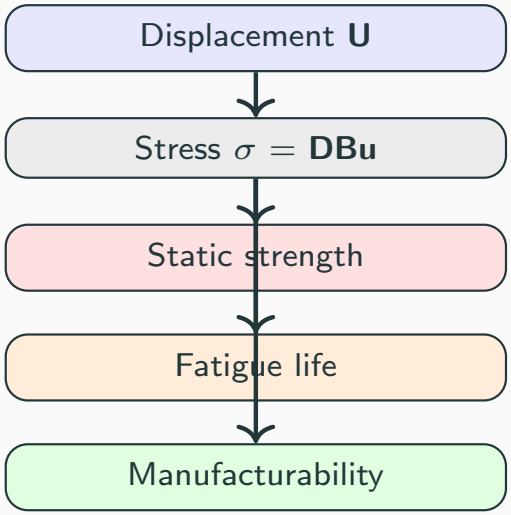
- **Fatigue constraint**: evaluate cyclic loading using a Goodman-type criterion **zhao2023topology**:

$$\frac{\sigma_a}{\sigma_e} + \frac{\sigma_m}{\sigma_{ut}} \leq 1$$

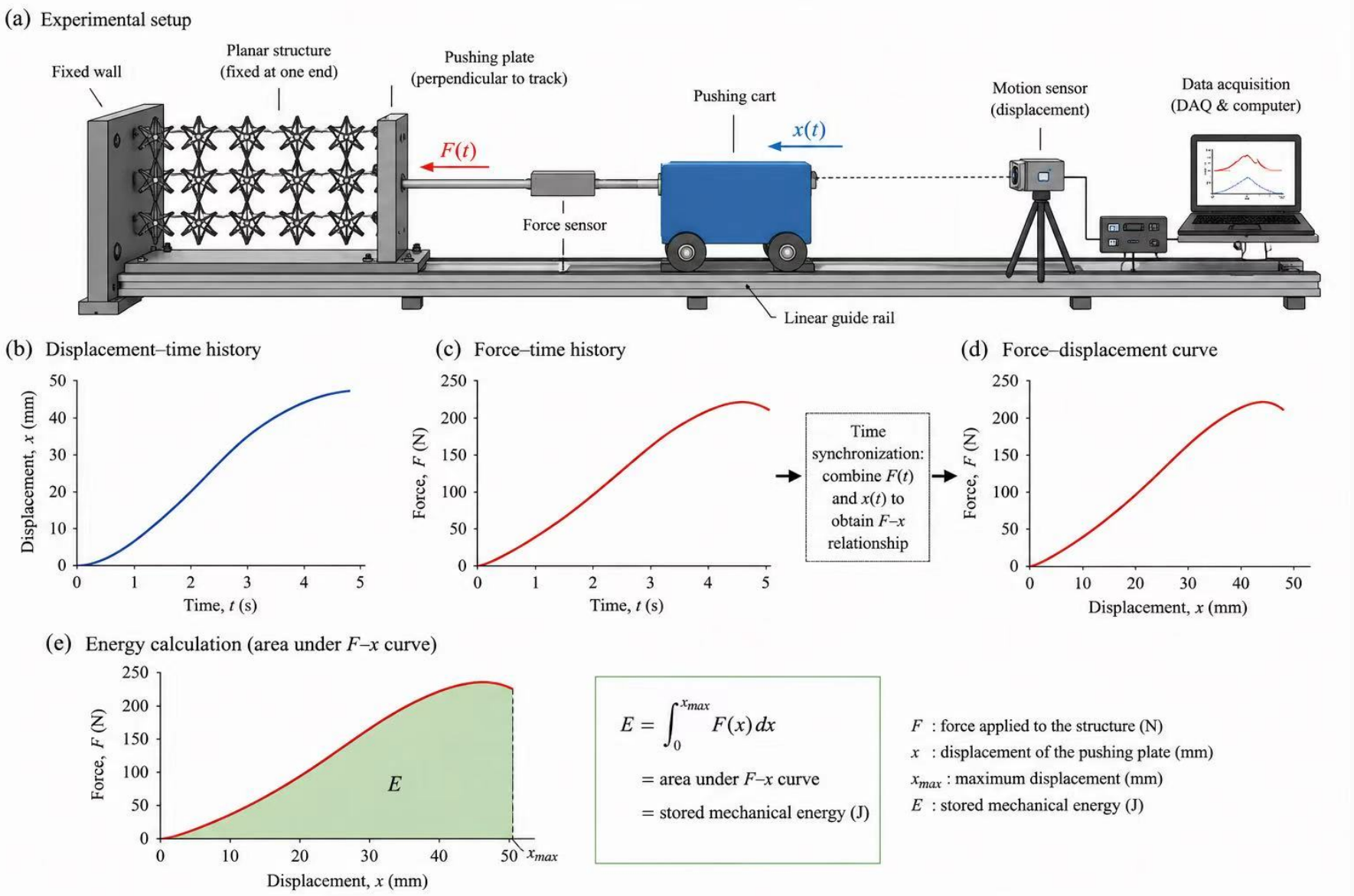
- Mean and alternating stresses are:

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}, \qquad \sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$$

- **Manufacturing**: density filtering



# Experimental Setup — Overview



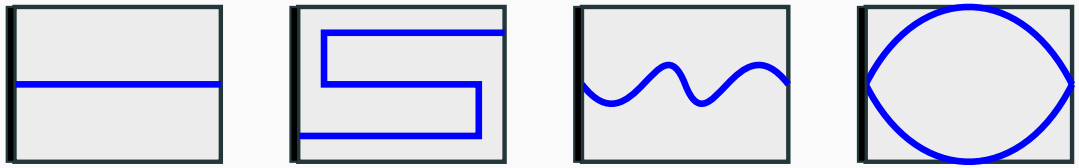
# Experimental Setup — Overview

- **Workflow:** Quasi-static compression $\rightarrow$ Time-synced $F(t)\&x(t)$ acquisition $\rightarrow$ $F - x$ relationship transformation for energy evaluation.
- **Loading Method:** Quasi-static manual pushing.
- **Data Acquisition:**
  - Displacement $x(t)$ recorded by Motion Sensor.
  - Applied Force $F(t)$ recorded by Force Sensor.
- **Stability:** High linearity in $x(t)$ indicates steady manual loading.

# Apparatus & Materials

- **Platform:** Hand wheel + lead screw compression rig (custom-built)
- **Force measurement:** Digital gauge, 0–50 N, 0.1 N resolution
- **Displacement:** Digital calliper, $\pm 0.01$ mm
- **Material:** PLA filament (1.75 mm), FDM 3D printing
- **Software:** GNU Octave, FreeCAD, Python
- **Environment:** Room temperature ($\sim 25\,^{\circ}\mathrm{C}$)
- **Material:** 3D printed PETG (1.75 mm); chosen for its superior toughness and interlayer adhesion compared to PLA, ensuring structural integrity during large deformations.

# Candidate Structures



(a) Straight (b) Folded (c) Serpentine (d) Curved

Four manual baselines + 1–2 topology-optimised candidates. All share identical footprint, thickness, and boundary conditions.

- **Design space:** 4 manual baselines vs. 1–2 topology-optimized candidates.
- **Constraints:** Identical footprint ($W \times L$), thickness, and PETG material.
- **Goal:** To evaluate the influence of geometric topology on energy storage $E$.

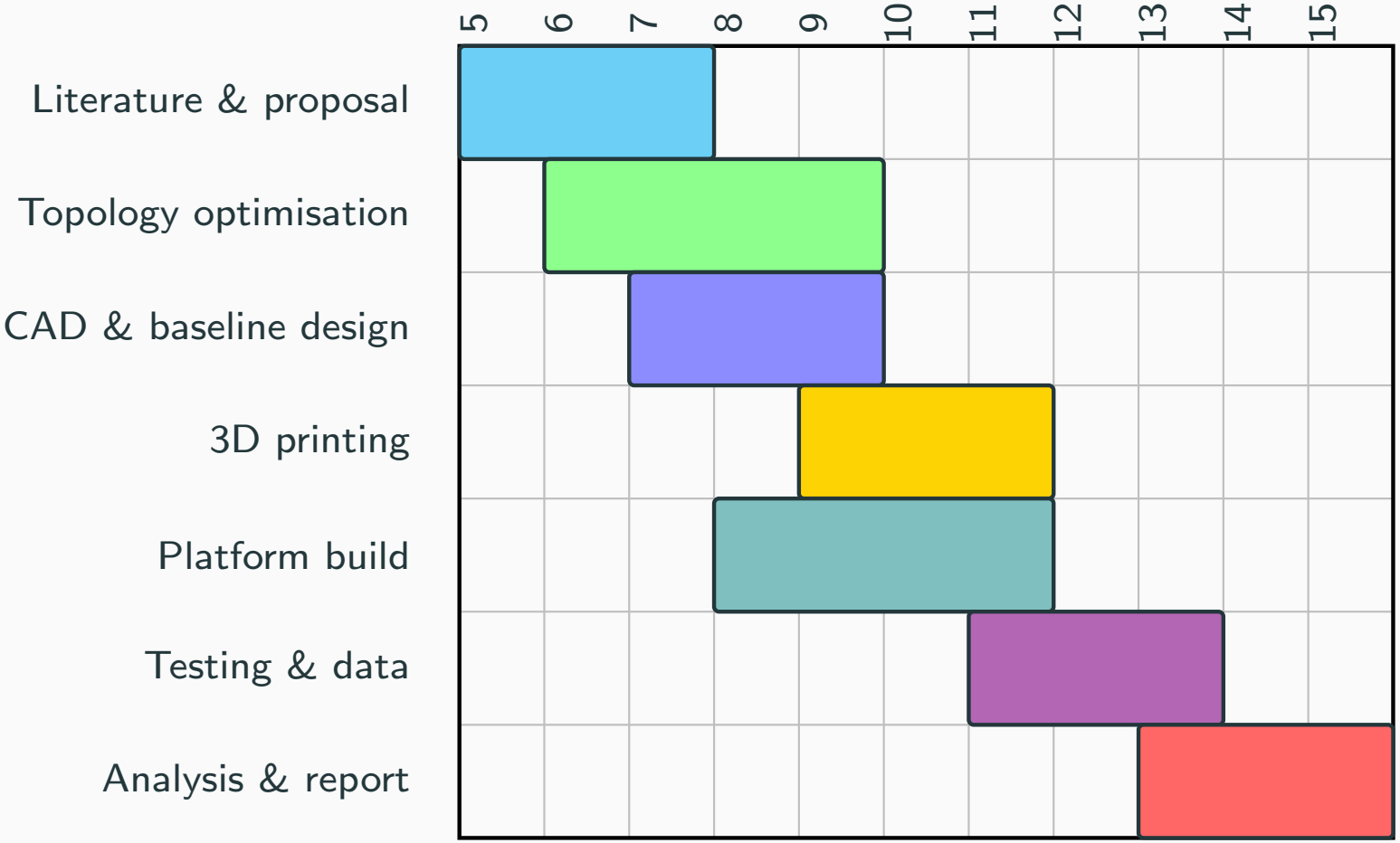
# Analysis Plan

1. **Data acquisition:** Record force $F$ and displacement $\delta$ at 0.1 mm increments during quasi-static compression. Repeat each topology $\geq 3$ times.
2. **Preprocessing:** Zero-offset correction; apply force sensor calibration curve; discard first cycle (pre-conditioning).
3. **Feature extraction:**
   - Stiffness: slope of linear region of $F$–$\delta$ curve.
   - $E_{\text{in}}$: area under loading curve (trapezoidal integration).
   - $E_{\text{out}}$: area under unloading curve.
   - $\eta = E_{\text{out}}/E_{\text{in}}$.
4. **Statistical treatment:** Mean $\pm$ standard deviation across trials; compare topologies using bar charts with error bars.
5. **Validation:** Compare experimental stiffness with FEA prediction (if time permits, COMSOL cross-check).

# Safety & Compliance

- **Hazards identified:**
  - 3D printer: hot nozzle ($\sim 200\,^{\circ}\mathrm{C}$), moving parts, minor fume emission from PLA.
  - Compression testing: spring-back of samples, pinch risk from lead screw mechanism.
- **Mitigation measures:**
  - Operate printer in ventilated area; do not touch nozzle during operation.
  - Wear safety glasses during compression tests.
  - Use hand wheel with controlled speed; never exceed elastic limit of samples.
- **Regulatory status:** No chemicals, radiation, or biological agents involved. Not regulated under Malaysian Occupational Safety and Health Act 1994.
- **Waste disposal:** PLA is biodegradable; failed prints collected in general recycling bin.
- **Emergency contacts:** Lab supervisor; campus security (ext. 8888).

# Project Timeline



# Budget

| Item | Qty | Unit (MYR) | Total (MYR) |
|---|---|---|---|
| *Equipment* | | | |
| Digital force gauge (0–50 N) | 1 | 45 | 45 |
| Lead screw + guide rail set | 1 | 35 | 35 |
| Hand wheel | 1 | 10 | 10 |
| *Materials & Consumables* | | | |
| PLA filament (1 kg spool) | 1 | 40 | 40 |
| Acrylic base plate | 1 | 15 | 15 |
| Bolts, nuts, clamps | – | – | 20 |
| *Software & Licences* | | | |
| Open-source (Octave, FreeCAD, Python) | – | – | 0 |
| *Miscellaneous* | | | |
| Spare filament & reprints | – | – | 20 |
| **Total** | | | **185** |

Requested from Physics Dept. student project fund (cap: MYR 500). 3D printer access via university makerspace.

# References i


[1] L. L. Howell, *Compliant Mechanisms*. New York: John Wiley & Sons, 2001.

[2] B. Zhu et al., "Design of compliant mechanisms using continuum topology optimization: A review," *Mechanism and Machine Theory*, vol. 143, p. 103 622, 2020. DOI: 10.1016/j.mechmachtheory.2019.103622

[3] X. Tong, Q. Ge, et al., "Design of planar springs for compliant energy storage mechanisms via topology optimization," *Journal of Mechanical Design*, vol. 145, no. 5, p. 051 701, 2023. DOI: 10.1115/1.4056959

[4] M. P. Bendsøe and O. Sigmund, *Topology Optimization: Theory, Methods, and Applications*. Berlin, Heidelberg: Springer, 2003. DOI: 10.1007/978-3-662-05086-6

# References ii


[5] O. Sigmund, "A 99 line topology optimization code written in Matlab," *Structural and Multidisciplinary Optimization*, vol. 21, no. 2, pp. 120–127, 2001. DOI: 10.1007/s001580050176

[6] E. Andreassen, A. Clausen, M. Schevenels, B. S. Lazarov, and O. Sigmund, "Efficient topology optimization in MATLAB using 88 lines of code," *Structural and Multidisciplinary Optimization*, vol. 43, no. 1, pp. 1–16, 2011. DOI: 10.1007/s00158-010-0594-7